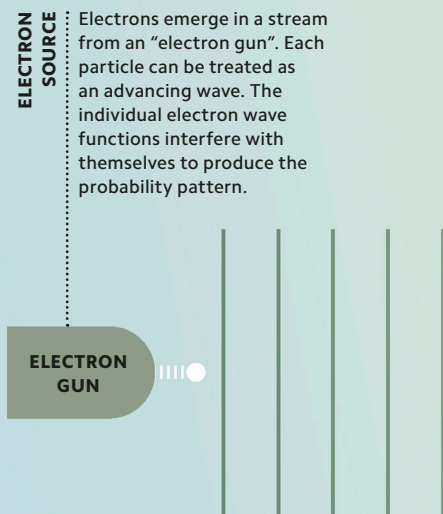


WAVE OR PARTICLE

At quantum scales, the dividing line between particles and waves (see pp.16–17) becomes blurred, with strange results. It is possible to design experiments that detect individual particlelike packets of energy, such as photons (quanta of electromagnetic radiation; see p.14), and at the same time demonstrate their wavelike behavior. Photons may arrive one at a time at a detector on the opposite side of two small slits, yet the pattern they build up can only be explained by each photon deciding on its location based upon wavelike interference (see pp.16–17).

Double slit experiment

A famous experiment conducted in 1800 to prove the wave nature of light can be adapted to show the wavelike nature of electrons and other particles.



The effect of measurements

One of the strangest aspects of quantum theory is that wave or particle behavior can be determined by the process of measurement.

PARTICLE DETECTOR

If we measure which slit each photon or electron passes through, they behave as particles at that slit and lose the wavelike behavior that existed prior to the slits.

